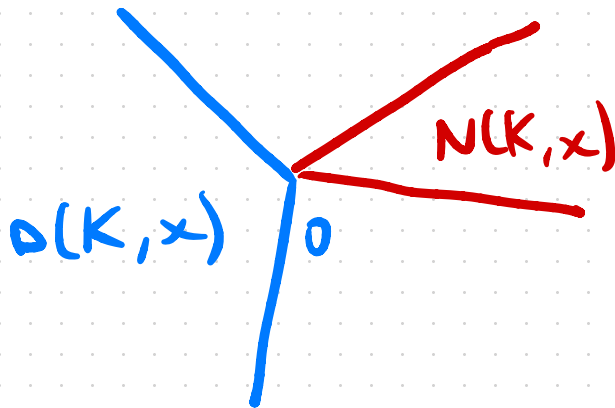
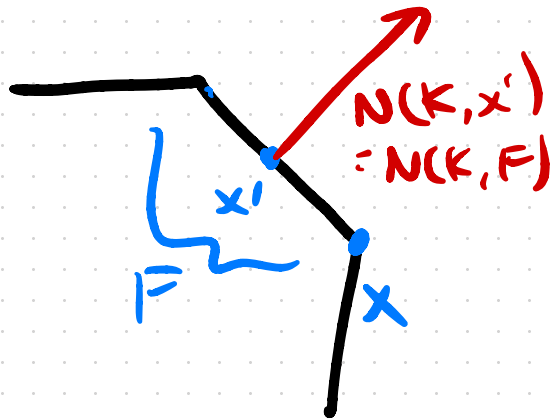


Let K be a convex set and $x \in K$. The normal cone of K at x is

$$N(K, x) := D(K, x)^\circ$$

Define $N(K, F) = N(K, x)$ for any $x \in \text{ri} F$.



Prop: Let K be a convex set, F a face of K , $x \in \text{rint } F$.

(1) TFAE:

(a) $y \in N(K, F)$

(b) $F \subset \arg\max_K y$

(c) $x \in \arg\max_K y$

(2) TFAE:

(a) $y \in \text{rint } N(K, F)$

(b) $F = \arg\max_K y$

$$P = \{x \in V : y_i(x) \leq b_i, i \in I\}, I \text{ finite.}$$

For any face F of P ,

$$F = P \cap \{x \in V : y_j(x) = b_j, j \in J\}$$

where J is the set of all $j \in I$ s.t. $y_j(x) = b_j$ for all $x \in F$.

$$D(P, F) = \{v \in V : y_j(v) \leq 0, j \in J\}$$

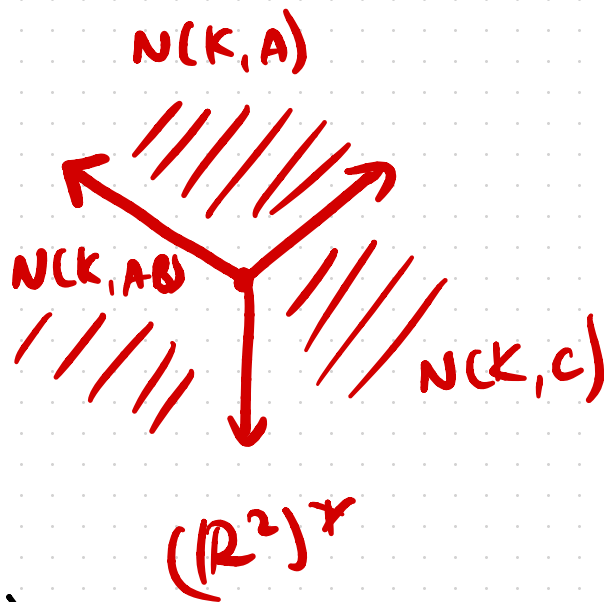
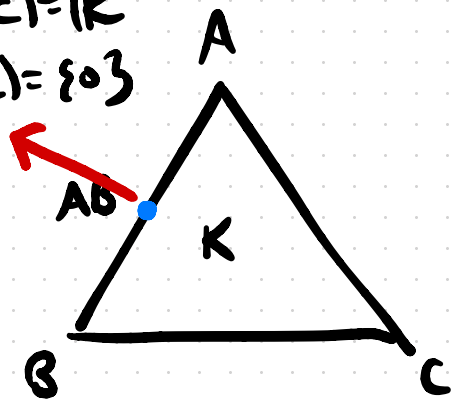
$$N(P, F) = \text{cone}(y_j : j \in J)$$

The **normal fan** $N(K)$ is the set of all normal cones of K .

$$N(K) = \{N(K, x) : x \in K\} = \{N(K, F) : F \in \mathcal{F}(K)\}$$

$$N(K, K) = \mathbb{R}^2$$

$$N(K, K) = \{0\}$$



$N(K)$ is a poset via inclusion

Theorem: Let P be a polyhedron.

(order-reversing)

(a) The posets $\mathcal{F}(P)$ and $\mathcal{N}(P)$ are anti-isomorphic, with anti-isomorphism $F \mapsto \mathcal{N}(P, F)$. Moreover,
 $\dim F + \dim \mathcal{N}(P, F) = \dim V$.

(b) If P is nonempty, then $\mathcal{N}(P)$ has unique minimal element $\mathcal{N}(P, P) = P^\perp$. $\leftarrow$ all linear functionals which are constant on P

(c) For any face F of P , the faces of $\mathcal{N}(P, F)$ are precisely the sets $\mathcal{N}(P, G)$ where G is a face of P containing F .

(d) If $P \neq \emptyset$,
$$\bigcup_{F \in \mathcal{F}(P)} \mathcal{N}(P, F) = (\text{recc } P)^\circ.$$

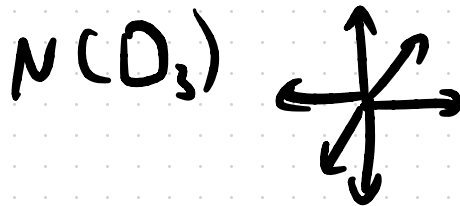
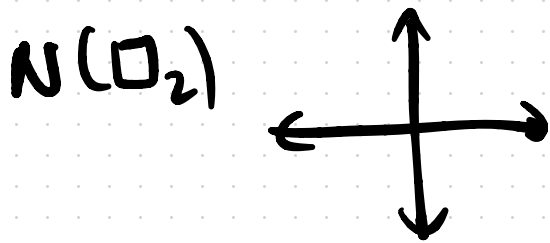
Example: $\square_n = \{x \in \mathbb{R}^n : -1 \leq x_i \leq 1 \ \forall i \in [n]\}$

Every face is of the form $\begin{matrix} e^i(x) \leq 1 \\ -e^i(x) \leq 1 \end{matrix}$

$$F = \square_n \cap \{x_i = 1, i \in S\} \cap \{x_i = -1, i \in T\}$$

$e^1, e^2, \dots, e^n$ standard dual basis

$$N(\square_n, F) = \text{cone}(e^i : i \in S, -e^i : i \in T)$$



$N(\square_n) =$
orthants of $\mathbb{R}^n$.

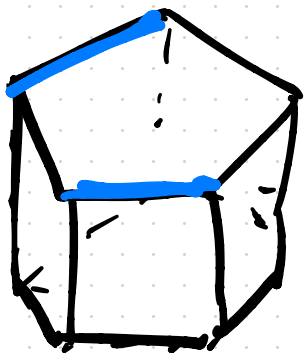
$F, G \in \mathcal{F}(P)$.

If F and G have nonempty intersection $F \cap G$, then $F \cap G$ is the greatest lower bound of F and G in $\mathcal{F}(P)$.

F and G also have a least upper bound:

$N(P, F), N(P, G)$ have a greatest lower bound $N(P, F) \cap N(P, G)$.

F and G have l.u.b. H , where $N(P, H) = N(P, F) \cap N(P, G)$.



For any convex set K , the support function of K is the function $h_K: V^* \rightarrow \mathbb{R} \cup \{\infty\}$

$$h_K(y) = \sup_{x \in K} y(x)$$

Prop: For any polyhedron P , h_P is a convex piecewise linear function on V^* , whose regions of linearity are given by $N(P)$.

